

Week 6 Submission Report

Student: Akshay

Phase: Phase 2 – Architectural Synthesis & Modular Building

Week: 6

Objective

Enhance the Product CRUD API with input validation, product search, and pagination while maintaining JWT authentication and modular architecture.

Tasks Completed

- Implemented product input validation for product name and price.
- Protected product endpoints with JWT authentication.
- Implemented product search by name using query parameters.
- Implemented product pagination using page, limit, SQL LIMIT and OFFSET.
- Updated Product Model with search and pagination database operations.
- Updated Product Controller with search and pagination logic.
- Updated Product Routes with search and pagination endpoints.
- Tested validation, search, pagination, and authentication using Thunder Client.
- Updated README.md with Week 6 API features.
- Created docs/week6.md documentation.

Week 6 API Features

GET /api/products/search?name=keyboard

GET /api/products/page?page=1&limit;=5

Validation Rules

Product name is required and product price must be a positive number. Invalid requests return HTTP 400 responses.

Testing Summary

The Week 6 features were successfully tested with Thunder Client. Valid requests were accepted, invalid product data was rejected, product search returned matching records, and pagination returned results according to the requested page and limit.

Conclusion

Week 6 development objectives were completed successfully. The backend now provides a more robust Product API with validation, search, pagination, and JWT-protected access.